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Texture Mapping Project Proposal

Goal: Build an interactive tool that maps 2D images and procedural shaders onto curved 3D surfaces (spheres, other smooth surfaces and scanned real-world meshes), and that measures and visualizes the distortion these mappings unavoidably introduce. As a stretch goal, we will extend the pipeline to the physical world by projecting the mapped content onto real objects with a calibrated projector, so it appears painted onto the object's surface.

Core Features:

1. **Mesh input:** load analytic surfaces (sphere, hemisphere, torus) and triangle meshes (test models and phone-scanned real objects), with basic cleanup for scanned meshes.
2. **Parameterization methods (our own implementations):**
    - Analytic sphere projections: equirectangular, Mercator, stereographic, cube map.
    - Tutte / harmonic embedding with a fixed circular boundary.
    - Least Squares Conformal Maps (LSCM) with a free boundary.
3. **Texturing and rendering:** apply images to meshes via the computed UV coordinates, plus 3D procedural shaders (noise, stripes, ripples, simulated lighting) that need no UV map.
4. **Distortion analysis:** per-triangle Jacobian and its singular values, giving angle distortion, area distortion, stretch and fold-over counts.
5. **Interactive viewer:**
    - Switch between parameterization methods.
    - Side-by-side 3D textured mesh and 2D UV layout.
    - Toggleable distortion heatmaps and checkerboard/circle-grid diagnostic textures.
6. **Quantitative evaluation:**
    - A methods and meshes distortion comparison.
    - Validation against analytic sphere-projection formulas.
    - Timing vs. mesh size.
    - A boundary/seam ablation.
    - Comparison against reference implementations.

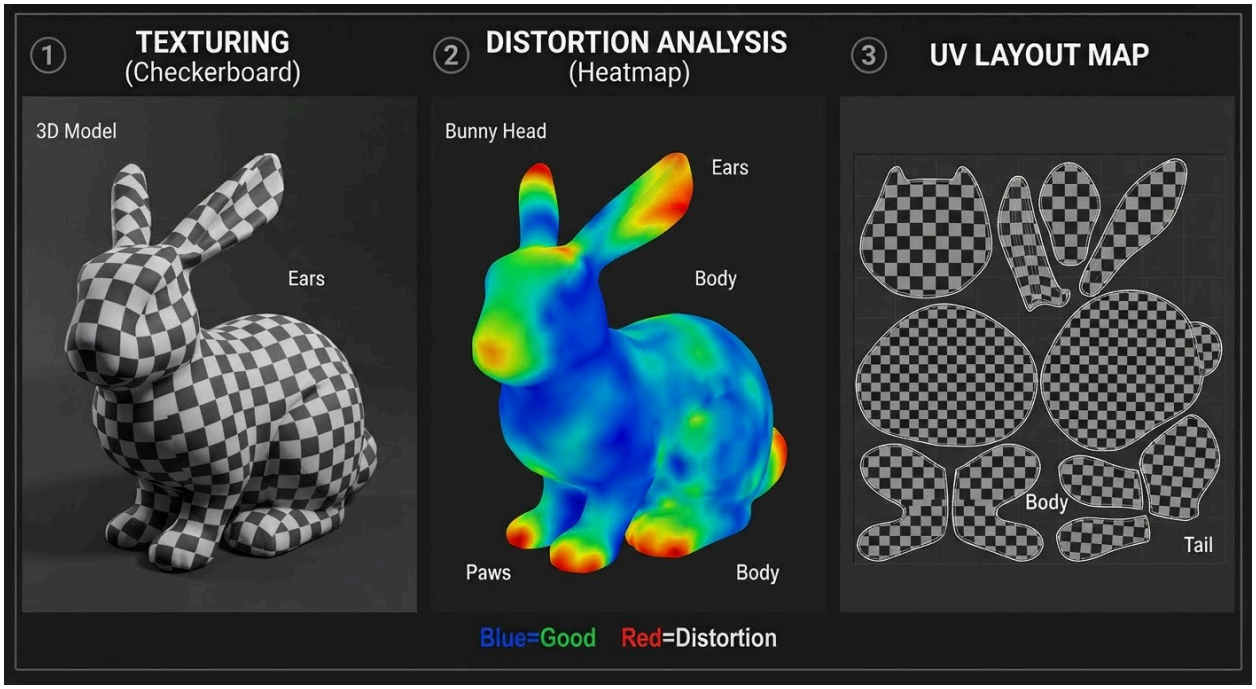
Stretch Features: .

- A1, projection onto a known shape: project onto a sphere or box with a real projector, with the pose estimated from manually selected correspondences, rendering from the projector's calibrated viewpoint.

- A2, projection onto a scanned object: scan an arbitrary object with phone photogrammetry, register it to the projector manually, UV-map it with our own code, and project.
- A3, structured-light scanning: scan the object using the projector and a camera with Gray-code patterns, producing a mesh already in the projector's coordinate frame for a fully automatic pipeline. Measure projection alignment error

Tools:
Python: Math and imaging pipelines
Polyscope: To visualize meshes and uv transforms
Git + Github: For version control
React: Possibly for interface
Three.js/WebGL: Rendering in web apps
GLSL: For shaders
Libgl : Python bindings, to validate our Tutte/LSCM/ARAP results, not as part of the submitted implementation.
Stretch goal tools: OpenCV (camera/projector calibration, structured-light decoding), a projector and webcam, and a phone photogrammetry app (e.g. Polycam or RealityScan) for scanning.

Mock Example: AI generated mock example for basic core functionality



Related Papers

Hormann, Lévy & Sheffer, ***Mesh Parameterization: Theory and Practice*** (SIGGRAPH 2007 course notes) https://www.inf.usi.ch/hormann/papers/Hormann.2008.MPT.pdf

Lévy, Petitjean, Ray & Maillot, ***Least Squares Conformal Maps for Automatic Texture Atlas Generation*** (SIGGRAPH 2002). https://inria.hal.science/inria-00100754/file/lscm.pdf

Sander, Snyder, Gortler & Hoppe, ***Texture Mapping Progressive Meshes*** (SIGGRAPH 2001). https://www.eecs.harvard.edu/~sjg/papers/tmpm.pdf

Raskar, Welch, Low & Bandyopadhyay, ***Shader Lamps: Animating Real Objects with Image-Based Illumination*** https://www.merl.com/publications/docs/TR2001-21.pdf